

Assignment 2: Secret Codes (Arrays & Functions)

Assignment for COSC 1100 – Intro to Programming

Prior to attempting this problem, you should have done the following:

1. View the Content, including assigned readings for weeks 5 through 9.
2. Viewed the Functions in Python video from week 9 and completed the associated Pre-Class Activity.

General Requirements

1. This assignment is to be completed **individually**. Group submissions will not be considered.
2. Analyze the problem, design a documented plan (i.e. flowchart or pseudo-code), code and test a solution **following the step-by-step approach** presented in this course.
3. Submit your solution and plan to the appropriate assignment folder on DC Connect **by the due date provided**.
4. Your instructor will assign a grade and post feedback on your submission to DC Connect.

Program Requirements

Create a simple Python application following the requirements below to allow a user to encode a piece of text using a cipher, or to decode the text using the same cipher. A “cipher” is a method of making a coded message that can be “deciphered” – usually the input to a cipher is the message to encode and some kind of “key” value that makes it make sense.

In your program, after the user picks whether they are encoding or decoding the text, the user will enter text and it will either be encoded or decoded. Since your program allows both encoding and decoding, you will be able to (*partly*) test if the cypher works using just your own program.

Detailed Requirements:

- **Initial prompt:** When the program starts the user is prompted to enter “1” to encode, “2” to decode, or “3” to exit.
- If the user inputs “1”:
 - Accept one text input from the user; validation of this input may not be necessary
 - Pass the user’s text input to an “encode” function
 - Print the encoded text string that was returned by the “encode” function
 - Do the **initial prompt** again
- If the user inputs “2”:
 - Accept one text input from the user; validation of this input may not be necessary
 - Pass the user’s text input to an “decode” function
 - Print the decoded text string that was returned by the “decode” function
 - Do the **initial prompt** again
- If the user inputs “3”:
 - End the program
- If the user inputs anything other than “1”, “2” or “3”, re-prompt them with the **initial prompt** again.

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Requirements for the Cipher Functions:

In this case, we will be using a cipher to translate a message to or from emoji. A [Substitution cipher](#) is a very simple type of cipher that operates by replacing characters or words from one form of communication with those from another. In our case, we will convert from English letters into emoji. Your encode and decode functions will each have one parameter, which is the message to be translated.

English	Unicode	Emoji
100 (100%)	U+1F4AF	
angry (mad)	U+1F620	
apple	U+1F34E	
banana	U+1F34C	
bird	U+1F426	
brain	U+1F9E0	
bus	U+1F68C	
car	U+1F697	
carrot	U+1F955	

English	Unicode	Emoji
cat	U+1F408	
clap	U+1F44F	
cold	U+1F976	
cookie	U+1F36A	
dog	U+1F415	
egg	U+1F95A	
fish	U+1F41F	
fox	U+1F98A	
happy	U+1F600	

English	Unicode	Emoji
heart	U+2764	
horse	U+1F40E	
hot	U+1F975	
map	U+1F5FA	
OK (okay)	U+1F44C	
sad	U+1F622	
sleep (bed)	U+1F6CC	
tree	U+1F332	
zombie	U+1F9DF	

For your assignment submission, please limit the words to translate to those included above, which were chosen intentionally by this course's professors. However, if you'd like to extend things somewhat, you can include the secondary words like (mad) and (okay) that are shown in parentheses in the chart.

There are many ways to implement this in Python, but probably the easiest way for our needs would be to declare two arrays: one includes a list of words, and the other includes the emoji equivalents of those words. Then, given a message, for each word, if that word is in the first array, match that word's position in the first array with the same position in the second array. Ensure that the translation happens regardless of case.

Here is some partial pseudocode for the encode function that you can make use of:

```
Start with an empty string called encoded_string
Use the .split() method to turn the input message into a list of words
For each word_index, counting up from 0 to the length of the input word list
    Assume that there is no match found (match_found = False)

    For each translation_index in the emoji word list
        If the word at word_index equals the input word at translation_index
            Add the emoji at translation_index to the encoded_string
            Set match_found to True
    If after going through the whole input list, match_found is False
        Add the original word at word_index to the encrypted string
```

Even perfectly translated into Python, the output from this may not be perfect and troubleshooting the differences in how the input displays is intentionally left as an exercise.

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Style Guide:

To be eligible for full marks on this or any assignment in this course your application must conform to the requirements as outlined above as well as our prescribed style guide, in this case making sure to observe the PEP8 naming conventions for Python as well as appropriate and complete program documentation.

Development Hints:

- As always, start with your planning steps. The pseudocode shown above may be part of the process and plan (steps 3 and 4 of 10).
- Only *after* planning, consider how you can “translate” the given pseudocode into Python code.
- Ciphers are a very common programming exercise and thus there are numerous reference materials and possibly even full solutions available on the web. However, be cautious with these!
 - Make sure any “borrowed” code *works* and that your program actually meets our requirements.
 - Make sure all code in your program conforms to our PEP8 style guide.
 - Reference any code directly taken from any form of online resource or another person, or this will be considered an incident of plagiarism. You can just do this in a comment before the part you have taken from another source.

Sample Output:

```
1) Encode text
2) Decode text
3) Exit
Please select an option between 1 and 3:1
Please enter the text to encode: My horse is angry. She will heart a carrot.
Encoded text: My 🐎 is 😡 She will ❤️ a 🥕

1) Encode text
2) Decode text
3) Exit
Please select an option between 1 and 3:3.1
Please select a valid option between 1 and 3.

1) Encode text
2) Decode text
3) Exit
Please select an option between 1 and 3:2
Please enter the text to decode: My 🐎 is 😡 She will ❤️ a 🥕
Decoded text: My horse is angry. She will heart a carrot.
```